

```
int main() {
    int nb, np;

    // Input number of blocks and processes
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    int block[nb], process[np], allocation[np];

    // Input block sizes
    printf("\nEnter size of each block:\n");
    for(int i = 0; i < nb; i++) {
        printf("Block %d: ", i + 1);
        scanf("%d", &block[i]);
    }

    // Input process sizes
    printf("\nEnter size of each process:\n");
    for(int i = 0; i < np; i++) {
        printf("Process %d: ", i + 1);
        scanf("%d", &process[i]);
        allocation[i] = -1; // Initially not allocated
    }

    // First Fit Allocation
    for(int i = 0; i < np; i++) {
        for(int j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                allocation[i] = j;
                block[j] -= process[i]; // reduce available size
                break;
            }
        }
    }
}
```

```
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ls
a.out backup fcf.c fcfs.c first.sh nithya.sh ntihya.sh prod.c roundro.c script.awk
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ vi firstfit.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ gcc firstfit.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ./a.out
```

Enter number of memory blocks: 6

Enter sizes of memory blocks:

20

40

15

5

20

40

Enter number of processes: 10

Enter sizes of processes:

14

24

25

13

17

4

10

37

18

13

Process No.	Process Size	Block No.
1	14	1
2	24	2
3	25	6
4	13	2
5	17	5
6	4	1
7	10	3
8	37	Not Allocated
9	18	Not Allocated
10	13	6

```
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$
```

```
3          25          6
4          13          2
5          17          5
6          4           1
7          10          3
8          37          Not Allocated
9          18          Not Allocated
10         13          6
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ls
a.out backup fcf.c fcfs.c first.sh firstfit.c nithya.sh ntihya.sh prod.c roundro.c script.awk
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ./a.out
```

Enter number of memory blocks: 6

Enter sizes of memory blocks:

1

20

15

19

30

16

Enter number of processes: 6

Enter sizes of processes:

15

20

25

45

23

18

Process No.	Process Size	Block No.
1	15	2
2	20	5
3	25	Not Allocated
4	45	Not Allocated
5	23	Not Allocated
6	18	4